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Machine Learning (CoSc3114) Lab Worksheet-4

Exercise on Supervised Machine Learning

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1. Apply 80-20 percentage split to partition Iris dataset to training and testing set.
2. Apply 10-fold cross-validation (10 CVD) to partition Iris dataset.
3. Apply stratified 10-fold cross-validation (10 CVD) to partition Iris dataset.
4. Consider the following table

Actual	P	P	N	N	P	N	N	N	P	N	N	N	N	P	N	P	P	P	P	N	P	N	N	P
Predicted	P	N	N	N	P	N	N	N	P	P	N	N	N	P	N	P	P	P	P	P	P	P	N	P

Where P stands for Positive Class Label

N stands for Positive Class Label

- ✓ Develop the confusion matrix
- ✓ Calculate the recall performance
- ✓ Calculate the precision performance value
- ✓ Calculate the Accuracy performance value
- ✓ Display the performance report that holds precision, recall, f1-score and accuracy of both Positive and Negative class label with Macro avg and weighted avg.
- ✓ Calculate the Area under the curve (AUC) using the TPR and FPR
- ✓ Draw the ROC curve using the TPR ad FPR

5. Consider the following table

Actual	1	0	1	0	0	0	1	1	1	0	1	0	0	0	1	1
Predicted	1	0	1.1	0.2	0.2	0	1.1	1.2	0.9	0.1	1	0	0	1	1	0.8

- ✓ Calculate the mean absolute error (MAE)
- ✓ Calculate the root mean squared error (RMSE)
- ✓ Calculate the squares error (MSE)

6. Write a python code to calculate the correlation of each features of iris dataset
7. Apply Decision tree algorithm using entropy (Information gain) to develop classifier model using 80/20 data split on IRIS dataset
   - ✓ Develop classifier model by splitting the dataset to 70-30 of training set and testing set respectively

- ✓ Generate the developed tree
- ✓ Represent the developed tree in terms of text
- ✓ Display the classification report that shows the performance of the classifier model tested by the testing dataset
- ✓ Generate the confusion matrix without class label
- ✓ Generate the confusion matrix by incorporating class label
- ✓ Calculate the accuracy, precision, recall and F-measure
- ✓ Find the class label of the record with the value of `{6.4, 1.8, 6.6, 2.1}`

8. Apply Decision tree using gini index to develop classifier model using 80/20 data split on IRIS dataset
   - ✓ Develop classifier model by splitting the dataset to 70-30 of training set and testing set respectively
   - ✓ Generate the developed tree
   - ✓ Represent the developed tree in terms of text
   - ✓ Display the classification report that shows the performance of the classifier model tested by the testing dataset
   - ✓ Generate the confusion matrix without class label
   - ✓ Generate the confusion matrix by incorporating class label
   - ✓ Calculate the accuracy, precision, recall and F-measure
   - ✓ Find the class label of the record with the value of `{6.4, 1.8, 6.6, 2.1}`
9. Apply Naïve Bayes algorithm (Categorical aive Bayes) on 'Buys Computer' dataset from lecture notes
   - ✓ Encode each values of the features to numerical values
   - ✓ Develop classifier model using the whole dataset as a training set
   - ✓ Predict the class label of the first 10 records
   - ✓ Evaluate the accuracy of the developed Naïve Bayes classifier on the first 10 records
10. Apply Naïve Bayes algorithm (Gaussian Naive Bayes) to develop classifier model using 70/30 data split on IRIS dataset
    - ✓ Split the dataset as training and testing
    - ✓ Develop classifier model by splitting the dataset to 70-30 of training set and testing set respectively
    - ✓ Develop classifier model using '`GaussianNB`'
    - ✓ Predict the class label of the testing dataset
    - ✓ Calculate the accuracy of the performance on the testing dataset
    - ✓ Predict the class label of the first record of the testing dataset
    - ✓ Calculate the accuracy of the performance on the first record of the testing dataset

11. Apply k-NN algorithm using ‘Euclidean distance’ on 70/30 data split on IRIS dataset
    - ✓ Split the dataset as 70/30 training and testing respectively
    - ✓ Develop classifier model using k-NN neighbor 5 and metric = ‘euclidean’
    - ✓ Predict the class label of the testing dataset
    - ✓ Calculate the accuracy of the performance on the testing dataset
    - ✓ Display the confusion matrix of the performance without labels
    - ✓ Display the confusion matrix of the performance with class labels
12. Repeat the k-NN algorithm using ‘Manhattan’ distance measure and 7 nearest neighbors
13. Apply feature selection using Information Gain ratio
    - ✓ Calculate the information gain of each feature of the iris dataset
    - ✓ Select the best 2 features of the iris dataset